Jacobian Derivation for a Hybrid 2-DoF Planar Cable-Driven Manipulator

CupManipulatorTendon — Kinematic & Dynamic Analysis

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Contents

1 System Description

The manipulator under consideration is a **hybrid 2-DoF planar serial arm** whose end-effector (EE) moves in a fixed horizontal plane (world $Z = z_0$). The two degrees of freedom are:

- q_1 (`link1_base`): revolute joint, *directly actuated* by motor torque τ_1.
- q_2 (`link2_link1`): revolute joint, *cable-driven* via a HTD 5M belt wound on a 60-tooth pulley. The controlled input is the cable tension $f_c \geq 0$.

This hybrid actuation is analogous to the Cable-Driven Parallel Manipulator (CDPM) framework presented in [1], where a structure matrix maps cable forces to generalised joint torques. Here, only joint 2 is governed by cable mechanics; joint 1 retains the classical direct-drive model.

Geometric parameters.

Symbol	Value	Description
L_1	342.47 mm	Distance from joint-1 axis to joint-2 axis (horizontal)
L_2	190.00 mm	Distance from joint-2 axis to EE origin (horizontal)
r_p	47.75 mm	HTD 5M 60T pulley pitch radius ($= 60 \times 5\,\text{mm}/(2\pi)$)
z_0	321.55 mm	Constant world-Z of the EE (set at home pose)

2 Coordinate Frames and Notation

Following the CDPM convention in the attached reference:

- $\{F_O\}$ — inertial (world) frame; origin at joint-1 axis.
- $\{F_\varepsilon\}$ — end-effector body frame; rigidly attached to link 2, origin at the EE point.
- Cable attachment on the *base* (drive pulley centre): ${}^{O}\boldsymbol{r}_{OA}$ — fixed in $\{F_O\}$.
- Cable attachment on the *EE link* (driven pulley winding point): ${}^{\varepsilon}\boldsymbol{r}_{OB}$ — fixed in $\{F_\varepsilon\}$.

Notation conventions: $c_i \triangleq \cos q_i,\;\; s_i \triangleq \sin q_i,\;\; c_{12} \triangleq \cos(q_1+q_2),\;\; s_{12} \triangleq \sin(q_1+q_2)$.

3 Forward Kinematics

The EE position in world frame $\{F_O\}$ is:

$$\boldsymbol{p}(q_1, q_2) = \begin{bmatrix} p_x \\ p_y \end{bmatrix} = \begin{bmatrix} L_1 c_1 + L_2 c_{12} \\ L_1 s_1 + L_2 s_{12} \end{bmatrix}, \tag{1}$$

where only the horizontal components are shown; $p_z = z_0 = \text{const}$.

4 Geometric (Translational) Jacobian

Differentiating eq. (1) with respect to $\boldsymbol{q} = [q_1, q_2]^\top$ gives the standard geometric Jacobian $\boldsymbol{J} \in \mathbb{R}^{2\times 2}$:

$$\boldsymbol{J}(\boldsymbol{q}) = \frac{\partial \boldsymbol{p}}{\partial \boldsymbol{q}} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}. \tag{2}$$

Velocity-level kinematics:

$$\dot{\boldsymbol{p}} = \boldsymbol{J}(\boldsymbol{q})\,\dot{\boldsymbol{q}}. \tag{3}$$

Singularity analysis. The determinant of $\boldsymbol{J}$ is:

$$\det \boldsymbol{J} = L_1 L_2 \sin q_2. \tag{4}$$

A kinematic singularity occurs whenever $q_2 = 0$ or $q_2 = \pm\pi$ (fully-extended or folded configurations). This is the fundamental reason that Jacobian-based (velocity) IK is unreliable near $q_2 = 0$ and analytical 2R IK must be used as a fallback.

5 Actuation Models

5.1 Joint 1 — Direct Actuation

Joint 1 is driven directly: the motor output torque τ_1 appears unmodified as the generalised force for q_1. There is no cable loop; the actuation ratio is unity.

5.2 Joint 2 — Cable (Belt/Pulley) Actuation

Following the CDPM cable-vector formulation [1], a single cable wraps around a drive pulley (radius r_p) mounted on the base and is anchored to a driven pulley fixed to link 2. The cable vector ${}^{\varepsilon}\boldsymbol{l}$ expressed in frame $\{F_\varepsilon\}$ is (cf. equation 3.7 in the reference):

$${}^{\varepsilon}\boldsymbol{l} = -{}^{\varepsilon}_{O}R\,{}^{O}\boldsymbol{r}_{OA} + {}^{\varepsilon}\boldsymbol{r}_{OB}, \tag{5}$$

where ${}^{\varepsilon}_{O}R$ is the rotation from $\{F_O\}$ to $\{F_\varepsilon\}$ and ${}^{O}\boldsymbol{r}_{OA}$ is the fixed world-frame position of the cable's base-side anchor. The cable length is $l = \|{}^{\varepsilon}\boldsymbol{l}\|$.

Belt–pulley simplification. Because both the drive and driven pulleys are of the same HTD-5M tooth profile and the belt is inextensible, the cable wraps without slip. The arc-length constraint gives a *linear* relationship between cable displacement Δl and joint angle increment Δq_2:

$$\dot{l} = r_p\,\dot{q}_2, \qquad \Delta l = r_p\,\Delta q_2. \tag{6}$$

This replaces the full CDPM cable-length formula: the wrapped geometry reduces to a scalar transmission ratio r_p.

Cable tension and joint torque. The cable tension $f_c \geq 0$ produces a torque about joint 2's axis via the moment arm r_p:

$$\tau_2 = r_p\,f_c. \tag{7}$$

6 Actuation Mapping Matrix

Collect the actuator inputs into the vector $\boldsymbol{u} = [\tau_1,\ f_c]^\top \in \mathbb{R}^2$ and the joint torques into $\boldsymbol{\tau} = [\tau_1,\ \tau_2]^\top$. From eq. (7):

$$\boldsymbol{\tau} = \boldsymbol{A}\,\boldsymbol{u}, \qquad \boldsymbol{A} = \begin{bmatrix} 1 & 0 \\ 0 & r_p \end{bmatrix}, \tag{8}$$

where $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ is the *actuation matrix*. Because $\boldsymbol{A}$ is diagonal and invertible (for $r_p > 0$), the system is fully actuated.

The inverse — mapping joint torques to actuator inputs — is:

$$\boldsymbol{A}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/r_p \end{bmatrix}. \tag{9}$$

7 Hybrid Jacobian Derivation

7.1 Task-Space Jacobian (Velocity Level)

From eqs. (3) and (6), the joint velocities $\dot{\boldsymbol{q}}$ relate to both the Cartesian EE velocity $\dot{\boldsymbol{p}}$ and the cable rate $\dot{l}$:

$$\begin{bmatrix} \dot{\boldsymbol{p}} \\ \dot{l} \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{J}(\boldsymbol{q}) & \\ \boldsymbol{0}^\top & r_p \end{bmatrix}}_{J_{\text{aug}} \in \mathbb{R}^{3\times 2}} \dot{\boldsymbol{q}}, \tag{10}$$

where $\boldsymbol{0}^\top = [0,\ 0]$ pads $\boldsymbol{J}$ to accommodate the cable row.

Explicitly:

$$J_{\text{aug}} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \\ 0 & r_p \end{bmatrix}. \tag{11}$$

7.2 Actuator-Space Jacobian

Define the *actuator velocity vector* $\dot{\boldsymbol{u}}_v = [\dot{q}_1,\ \dot{l}]^\top$ (motor angular rate and cable linear rate). The joint velocities are recovered via $\boldsymbol{A}^{-1}$:

$$\dot{\boldsymbol{q}} = \boldsymbol{A}^{-1}\,\dot{\boldsymbol{u}}_v = \begin{bmatrix} 1 & 0 \\ 0 & 1/r_p \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{l} \end{bmatrix}. \tag{12}$$

Substituting into eq. (3):

$$\dot{\boldsymbol{p}} = \boldsymbol{J}(\boldsymbol{q})\,\boldsymbol{A}^{-1}\,\dot{\boldsymbol{u}}_v = \underbrace{\boldsymbol{J}(\boldsymbol{q})\,\boldsymbol{A}^{-1}}_{\boldsymbol{J}_h(\boldsymbol{q})}\,\dot{\boldsymbol{u}}_v, \tag{13}$$

$$\boxed{\boldsymbol{J}_h(\boldsymbol{q}) \;=\; \boldsymbol{J}(\boldsymbol{q})\,\boldsymbol{A}^{-1} \;=\; \begin{bmatrix} -(L_1 s_1 + L_2 s_{12}) & -\dfrac{L_2 s_{12}}{r_p} \\ L_1 c_1 + L_2 c_{12} & \dfrac{L_2 c_{12}}{r_p} \end{bmatrix}} \tag{14}$$

$\boldsymbol{J}_h \in \mathbb{R}^{2\times 2}$ is the **hybrid Jacobian**: it maps the *physical actuator rates* (joint-1 angular velocity, cable linear velocity) to the Cartesian EE velocity.

Interpretation. The first column of $\boldsymbol{J}_h$ is identical to the first column of $\boldsymbol{J}$: rotating joint 1 at rate $\dot{q}_1$ sweeps the EE through the same arc regardless of the drive type. The second column is the standard cable column scaled by $1/r_p$, converting from cable linear speed to the equivalent joint angular speed.

7.3 Singularity of the Hybrid Jacobian

$$\det \boldsymbol{J}_h = \frac{1}{r_p} \det \boldsymbol{J} = \frac{L_1 L_2}{r_p} \sin q_2. \tag{15}$$

Since $r_p > 0$, $\boldsymbol{J}_h$ shares the same singularity locus as the geometric Jacobian: $q_2 = 0 \pmod{\pi}$.

8 Cable-Force Jacobian and Statics

Following the CDPM structure-matrix formulation (equation 3.15 in the reference), the transpose of the Jacobian maps Cartesian EE forces to actuator inputs via virtual work:

$$\delta W = \boldsymbol{F}_{EE}^\top \delta \boldsymbol{p} = \boldsymbol{\tau}^\top \delta \boldsymbol{q} = \boldsymbol{u}^\top \boldsymbol{A}^\top \delta \boldsymbol{q}. \tag{16}$$

Since $\delta \boldsymbol{p} = \boldsymbol{J}\, \delta \boldsymbol{q}$:

$$\boldsymbol{F}_{EE}^\top \boldsymbol{J}\, \delta \boldsymbol{q} = \boldsymbol{u}^\top \boldsymbol{A}^\top \delta \boldsymbol{q} \implies \boldsymbol{J}^\top \boldsymbol{F}_{EE} = \boldsymbol{A}^\top \boldsymbol{u}, \tag{17}$$

$$\boldsymbol{u} = (\boldsymbol{A}^\top)^{-1} \boldsymbol{J}^\top \boldsymbol{F}_{EE} = \boldsymbol{J}_h^\top \boldsymbol{F}_{EE}. \tag{18}$$

Explicitly:

$$\boldsymbol{J}_h^\top = \begin{bmatrix} -(L_1 s_1 + L_2 s_{12}) & L_1 c_1 + L_2 c_{12} \\ -\dfrac{L_2 s_{12}}{r_p} & \dfrac{L_2 c_{12}}{r_p} \end{bmatrix}. \tag{19}$$

The first row gives the required motor torque τ_1; the second row gives the required cable tension f_c to produce the desired EE wrench $\boldsymbol{F}_{EE}$.

9 Velocity-Level Inverse Kinematics

9.1 Standard Damped Pseudo-Inverse on $\boldsymbol{J}$

Given a desired EE velocity $\dot{\boldsymbol{p}}_{\text{des}}$, the joint velocity is:

$$\dot{\boldsymbol{q}} = \boldsymbol{J}^\dagger \dot{\boldsymbol{p}}_{\text{des}} = \boldsymbol{J}^\top \left(\boldsymbol{J}\boldsymbol{J}^\top + \lambda \boldsymbol{I}_2\right)^{-1} \dot{\boldsymbol{p}}_{\text{des}}, \tag{20}$$

where $\lambda > 0$ is the damping factor (Levenberg–Marquardt regularisation). This is the method implemented in `compute_velocity_ik`.

9.2 Hybrid-Jacobian Velocity IK

To work directly in actuator space (i.e., command motor velocity $\dot{q}_1$ and cable rate $\dot{l}$):

$$\dot{\boldsymbol{u}}_v = \boldsymbol{J}_h^\dagger \dot{\boldsymbol{p}}_{\text{des}} = \boldsymbol{J}_h^\top \left(\boldsymbol{J}_h \boldsymbol{J}_h^\top + \lambda \boldsymbol{I}_2\right)^{-1} \dot{\boldsymbol{p}}_{\text{des}}. \tag{21}$$

Then the cable command is $\dot{l}$ and the motor-1 command is $\dot{q}_1 = [\dot{\boldsymbol{u}}_v]_1$.

9.3 Equivalence

Because $\boldsymbol{A}^{-1}$ is invertible:

$$\boldsymbol{J}_h^{\dagger} = \boldsymbol{A}^{-1}\boldsymbol{J}^{\dagger} \quad \text{(when } \boldsymbol{A} \text{ is invertible)}. \tag{22}$$

The two velocity-IK expressions eqs. (20) and (21) yield the same EE motion; they differ only in which actuator space the solution is expressed.

10 Equations of Motion

The standard manipulator dynamics in joint space:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{\tau} + \boldsymbol{J}^{\top}\boldsymbol{F}_{\text{ext}}, \tag{23}$$

where $\boldsymbol{M}$ is the mass-inertia matrix, $\boldsymbol{C}$ the Coriolis/centrifugal matrix, and $\boldsymbol{G}$ the gravitational vector. Substituting $\boldsymbol{\tau} = \boldsymbol{A}\,\boldsymbol{u}$ from eq. (8):

$$\boxed{\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{A}\,\boldsymbol{u} + \boldsymbol{J}^{\top}\boldsymbol{F}_{\text{ext}}.} \tag{24}$$

For torque control, the desired actuator inputs are computed as:

$$\boldsymbol{u} = \boldsymbol{A}^{-1}\big(\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\text{ref}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_{\text{ref}} + \boldsymbol{G}(\boldsymbol{q}) - \boldsymbol{J}^{\top}\boldsymbol{F}_{\text{ext}}\big). \tag{25}$$

11 Summary

Symbol	Expression	Role
$\boldsymbol{J}(\boldsymbol{q})$	eq. (2)	Geometric Jacobian (2×2)
$\boldsymbol{A}$	eq. (8)	Actuation matrix (diagonal, 2×2)
$\boldsymbol{J}_h = \boldsymbol{J}\boldsymbol{A}^{-1}$	eq. (14)	Hybrid Jacobian (actuator $\to$ task)
$\boldsymbol{J}_h^{\top}$	eq. (19)	Maps EE wrench $\to$ actuator inputs
J_{aug}	eq. (11)	Augmented Jacobian (task + cable, 3×2)
$\dot{\boldsymbol{u}}_v = \boldsymbol{J}_h^{\dagger}\,\dot{\boldsymbol{p}}$	eq. (21)	Hybrid velocity IK

Key relationships at a glance:

$$\begin{aligned}
\dot{\boldsymbol{p}} &= \boldsymbol{J}_h\,\dot{\boldsymbol{u}}_v && \text{(actuator velocity} \to \text{EE velocity)}\\
\boldsymbol{u} &= \boldsymbol{J}_h^{\top}\boldsymbol{F}_{EE} && \text{(EE force} \to \text{actuator inputs, static)}\\
\dot{l} &= r_p\,\dot{q}_2 && \text{(cable rate from joint-2 rate)}\\
\det \boldsymbol{J}_h &= (L_1 L_2/r_p)\sin q_2 && \text{(singular at } q_2 = 0, \pm\pi)
\end{aligned}$$

12 Cable Kinematics via Tangency Cancellation

This section derives the cable length rate $\dot{l}$ rigorously for a cable path that wraps around the driven-pulley body before attaching to the load point. The key result is that the velocity of the intermediate tangency points cancels, leaving a single clean expression involving only the load-attachment point.

12.1 Assumptions and Setup

Assume the driven-pulley centre O_2 is *fixed in the world*:

$$ {}^{W}\mathbf{r}_{WO_2} = \text{constant} \implies {}^{W}\mathbf{v}_{O_2} = {}^{W}\dot{\mathbf{r}}_{WO_2} = \mathbf{0}. \tag{26} $$

The variable cable path is

$$ B_{0,L} \;\rightarrow\; A_{1,L} \;\rightsquigarrow\; B_{1,L} \;\rightarrow\; A_{2,R}, $$

where:

- $B_{0,L}$ — fixed anchor point in the world frame.
- $A_{1,L}$ — tangency point where the cable *leaves* the straight segment and begins wrapping on the pulley.
- $B_{1,L}$ — tangency point where the cable *leaves* the pulley and begins the final straight segment.
- $A_{2,R}$ — load-attachment point, rigidly fixed to the rotating body.

12.2 Decomposition of Total Cable Length

The total cable length is:

$$ l = l_{B_{0,L}A_{1,L}} + l_{A_{1,L}B_{1,L}} + l_{B_{1,L}A_{2,R}}. \tag{27} $$

Differentiating:

$$ \dot{l} = \dot{l}_{B_{0,L}A_{1,L}} + \dot{l}_{A_{1,L}B_{1,L}} + \dot{l}_{B_{1,L}A_{2,R}}. \tag{28} $$

12.3 First Straight Segment

Define ${}^{W}\mathbf{s}_1 = {}^{W}\mathbf{r}_{WA_{1,L}} - {}^{W}\mathbf{r}_{WB_{0,L}}$. Using $\frac{d}{dt}\|\mathbf{x}\| = \frac{\mathbf{x}^\top\dot{\mathbf{x}}}{\|\mathbf{x}\|}$ and noting that $B_{0,L}$ is fixed:

$$ \dot{l}_{B_{0,L}A_{1,L}} = {}^{W}\hat{\mathbf{u}}^\top_{B_{0,L}A_{1,L}}\,{}^{W}\dot{\mathbf{r}}_{WA_{1,L}}, \qquad {}^{W}\hat{\mathbf{u}}_{B_{0,L}A_{1,L}} = \frac{{}^{W}\mathbf{r}_{WA_{1,L}} - {}^{W}\mathbf{r}_{WB_{0,L}}}{\left\|{}^{W}\mathbf{r}_{WA_{1,L}} - {}^{W}\mathbf{r}_{WB_{0,L}}\right\|}. \tag{29} $$

12.4 Wrapped Arc

For the arc $A_{1,L} \to B_{1,L}$, define unit tangents ${}^{W}\mathbf{t}_{A_{1,L}}$ and ${}^{W}\mathbf{t}_{B_{1,L}}$ pointing *outward from the arc* at each endpoint (i.e. opposite to the direction of cable travel). Then:

$$ \dot{l}_{A_{1,L}B_{1,L}} = +{}^{W}\mathbf{t}^\top_{A_{1,L}}\,{}^{W}\dot{\mathbf{r}}_{WA_{1,L}} - {}^{W}\mathbf{t}^\top_{B_{1,L}}\,{}^{W}\dot{\mathbf{r}}_{WB_{1,L}}. \tag{30} $$

Note on sign convention. With the standard incoming-tangent convention ($\mathbf{t}$ pointing in the direction of travel), the signs would be reversed. The outward convention used here ensures the tangency cancellation in section 12.7 follows with the same sign as the straight-segment formula.

12.5 Last Straight Segment

Define ${}^{W}\mathbf{s}_2 = {}^{W}\mathbf{r}_{WA_{2,R}} - {}^{W}\mathbf{r}_{WB_{1,L}}$:

$$\dot{l}_{B_{1,L}A_{2,R}} = {}^{W}\hat{\mathbf{u}}^{\top}_{B_{1,L}A_{2,R}}\left({}^{W}\dot{\mathbf{r}}_{WA_{2,R}} - {}^{W}\dot{\mathbf{r}}_{WB_{1,L}}\right), \quad {}^{W}\hat{\mathbf{u}}_{B_{1,L}A_{2,R}} = \frac{{}^{W}\mathbf{r}_{WA_{2,R}} - {}^{W}\mathbf{r}_{WB_{1,L}}}{\left\|{}^{W}\mathbf{r}_{WA_{2,R}} - {}^{W}\mathbf{r}_{WB_{1,L}}\right\|}. \tag{31}$$

12.6 Summing All Three Contributions

Adding the three parts:

$$\begin{aligned}\dot{l} = {} & \underbrace{\left({}^{W}\hat{\mathbf{u}}_{B_{0,L}A_{1,L}} + {}^{W}\mathbf{t}_{A_{1,L}}\right)^{\top}}_{\text{coefficient of } \dot{\mathbf{r}}_{WA_{1,L}}} {}^{W}\dot{\mathbf{r}}_{WA_{1,L}} + {}^{W}\hat{\mathbf{u}}^{\top}_{B_{1,L}A_{2,R}} {}^{W}\dot{\mathbf{r}}_{WA_{2,R}} \\ & - \underbrace{\left({}^{W}\mathbf{t}_{B_{1,L}} + {}^{W}\hat{\mathbf{u}}_{B_{1,L}A_{2,R}}\right)^{\top}}_{\text{coefficient of } {}^{W}\dot{\mathbf{r}}_{WB_{1,L}}} {}^{W}\dot{\mathbf{r}}_{WB_{1,L}}.\end{aligned} \tag{32}$$

12.7 Tangency Cancellation

At each tangency point the cable direction (straight segment unit vector) must equal the outward arc tangent for the cable to be smooth (no kink). Therefore:

$${}^{W}\mathbf{t}_{A_{1,L}} = -{}^{W}\hat{\mathbf{u}}_{B_{0,L}A_{1,L}}, \qquad {}^{W}\mathbf{t}_{B_{1,L}} = -{}^{W}\hat{\mathbf{u}}_{B_{1,L}A_{2,R}}. \tag{33}$$

Both bracket terms vanish identically:

$${}^{W}\hat{\mathbf{u}}_{B_{0,L}A_{1,L}} + {}^{W}\mathbf{t}_{A_{1,L}} = \mathbf{0}, \qquad {}^{W}\mathbf{t}_{B_{1,L}} + {}^{W}\hat{\mathbf{u}}_{B_{1,L}A_{2,R}} = \mathbf{0}. \tag{34}$$

The velocities of the two tangency points $A_{1,L}$ and $B_{1,L}$ drop out completely, yielding:

$$\boxed{\dot{l} = {}^{W}\hat{\mathbf{u}}^{\top}_{B_{1,L}A_{2,R}} \; {}^{W}\dot{\mathbf{r}}_{WA_{2,R}}} \tag{35}$$

12.8 Rigid-Body Velocity of the Load-Attachment Point

Because $A_{2,R}$ is fixed to the body rotating about the fixed point O_2, and ${}^{W}\dot{\mathbf{r}}_{WO_2} = \mathbf{0}$:

$${}^{W}\dot{\mathbf{r}}_{WA_{2,R}} = {}^{W}\boldsymbol{\omega}_2 \times {}^{W}\mathbf{r}_{O_2A_{2,R}}. \tag{36}$$

Substituting into eq. (35):

$$\boxed{\dot{l} = {}^{W}\hat{\mathbf{u}}^{\top}_{B_{1,L}A_{2,R}}\left({}^{W}\boldsymbol{\omega}_2 \times {}^{W}\mathbf{r}_{O_2A_{2,R}}\right)} \tag{37}$$

12.9 Planar Reduction

For planar motion, ${}^{W}\boldsymbol{\omega}_2 = [0,\ 0,\ \dot{\theta}_2]^{\top}$, giving:

$$\boxed{\dot{l} = {}^{W}\hat{\mathbf{u}}^{\top}_{B_{1,L}A_{2,R}}\left(\begin{bmatrix}0\\0\\\dot{\theta}_2\end{bmatrix} \times {}^{W}\mathbf{r}_{O_2A_{2,R}}\right) \;=\; J_l(\theta_2)\,\dot{\theta}_2.} \tag{38}$$

The scalar Jacobian entry $J_l(\theta_2)$ depends only on the current geometry of the cable's last straight segment and the lever arm ${}^{W}\mathbf{r}_{O_2A_{2,R}}$; it replaces the constant r_p approximation used in the simplified model of section 5.2. When the cable wrapping is symmetric and $A_{2,R}$ lies on the pitch circle, $J_l = r_p$, recovering the belt formula eq. (6).

12.10 Spatial-Velocity Row Jacobian Form

The result eq. (37) can be rewritten in the compact *spatial velocity* form that appears in cable-robot literature, making the translational and rotational contributions explicit side-by-side.

Triple-product identity. For any vectors $\mathbf{u}$, $\boldsymbol{\omega}$, $\mathbf{r}$:

$$\mathbf{u}^\top(\boldsymbol{\omega} \times \mathbf{r}) = (\mathbf{r} \times \mathbf{u})^\top \boldsymbol{\omega}. \tag{39}$$

Apply this to eq. (37) with $\mathbf{u} \triangleq {}^W\hat{\mathbf{u}}_{B_{1,L}A_{2,R}}$ and $\mathbf{r} \triangleq {}^W\mathbf{r}_{O_2A_{2,R}}$:

$$\dot{l} = \mathbf{u}^\top \underbrace{{}^W\mathbf{v}_{O_2}}_{\mathbf{0}} + \mathbf{u}^\top({}^W\boldsymbol{\omega}_2 \times \mathbf{r}) = \mathbf{u}^\top\, {}^W\mathbf{v}_{O_2} + (\mathbf{r} \times \mathbf{u})^\top\, {}^W\boldsymbol{\omega}_2. \tag{40}$$

Writing this as a single row times a spatial velocity vector:

$$\boxed{\dot{l} = \underbrace{\begin{bmatrix} {}^W\hat{\mathbf{u}}^\top_{B_{1,L}A_{2,R}} & \left({}^W\mathbf{r}_{O_2A_{2,R}} \times {}^W\hat{\mathbf{u}}_{B_{1,L}A_{2,R}}\right)^\top \end{bmatrix}}_{V \in \mathbb{R}^{1\times 6}} \begin{bmatrix} {}^W\mathbf{v}_{O_2} \\ {}^W\boldsymbol{\omega}_2 \end{bmatrix}} \tag{41}$$

The first block ${}^W\hat{\mathbf{u}}^\top \in \mathbb{R}^{1\times 3}$ captures the *translational* contribution; the second block $(\mathbf{r} \times \mathbf{u})^\top \in \mathbb{R}^{1\times 3}$ captures the *rotational* (moment-arm) contribution. This is structurally identical to the structure matrix row in cable-driven parallel robot formulations.

Fixed-pivot reduction. Since O_2 is fixed, ${}^W\mathbf{v}_{O_2} = \mathbf{0}$, and eq. (41) reduces to:

$$\boxed{\dot{l} = \left({}^W\mathbf{r}_{O_2A_{2,R}} \times {}^W\hat{\mathbf{u}}_{B_{1,L}A_{2,R}}\right)^\top {}^W\boldsymbol{\omega}_2} \tag{42}$$

This is the rotational-only form: the cable rate equals the moment-arm vector (lever arm crossed with cable unit vector) dotted with the angular velocity of the driven link.

12.11 Multi-Cable Structure Matrix

For a system with m cables, each cable i contributes one row. Stack all rows to obtain the *structure matrix* $V \in \mathbb{R}^{m\times 6}$:

$$\boxed{V = \begin{bmatrix} {}^W\hat{\mathbf{u}}_1^\top & \left({}^W\mathbf{r}_1 \times {}^W\hat{\mathbf{u}}_1\right)^\top \\ {}^W\hat{\mathbf{u}}_2^\top & \left({}^W\mathbf{r}_2 \times {}^W\hat{\mathbf{u}}_2\right)^\top \\ \vdots & \vdots \\ {}^W\hat{\mathbf{u}}_m^\top & \left({}^W\mathbf{r}_m \times {}^W\hat{\mathbf{u}}_m\right)^\top \end{bmatrix}, \qquad \dot{\mathbf{l}} = V \begin{bmatrix} {}^W\mathbf{v} \\ {}^W\boldsymbol{\omega} \end{bmatrix},} \tag{43}$$

where ${}^W\hat{\mathbf{u}}_i \triangleq {}^W\hat{\mathbf{u}}_{B^{(i)}_{1,L}A^{(i)}_{2,R}}$ and ${}^W\mathbf{r}_i \triangleq {}^W\mathbf{r}_{O_2A^{(i)}_{2,R}}$ for the i-th cable, and the spatial velocity $[{}^W\mathbf{v}^\top,\ {}^W\boldsymbol{\omega}^\top]^\top$ is evaluated at the pivot point O_2.

Connection to the hybrid Jacobian. For the single cable driving joint 2 of the planar arm the angular velocity is ${}^W\boldsymbol{\omega}_2 = \dot{q}_2\,\hat{\mathbf{k}}$ with $\hat{\mathbf{k}} = [0, 0, 1]^\top$. Substituting into eq. (42):

$$\dot{l} = \left({}^W\mathbf{r}_{O_2A_{2,R}} \times {}^W\hat{\mathbf{u}}_{B_{1,L}A_{2,R}}\right)^\top \hat{\mathbf{k}}\,\dot{q}_2 = J_l(q_2)\,\dot{q}_2, \tag{44}$$

recovering precisely the scalar Jacobian entry from eq. (38).

Derivation of the belt formula $J_l = r_p$. We now show explicitly that the tangency geometry forces $J_l = r_p$.

Setup. Let ϕ be the angle that the lever arm makes with the world x-axis at an arbitrary cable configuration. With $A_{2,R}$ on the pitch circle of radius r_p:

$$
{}^{W}\mathbf{r}_{O_2A_{2,R}} = r_p \begin{bmatrix} \cos\phi \\ \sin\phi \\ 0 \end{bmatrix}.
$$

Tangency condition. Because the cable departs the pulley tangentially, the cable direction must be perpendicular to the lever arm: ${}^{W}\hat{\mathbf{u}} \perp {}^{W}\mathbf{r}_{O_2A_{2,R}}$, i.e. the angle between them is 90. The unique planar unit tangent is:

$$
{}^{W}\hat{\mathbf{u}}_{B_{1,L}A_{2,R}} = \begin{bmatrix} -\sin\phi \\ \cos\phi \\ 0 \end{bmatrix} \qquad \text{(or its negation, depending on winding direction).}
$$

Cross product.

$$
{}^{W}\mathbf{r}_{O_2A_{2,R}} \times {}^{W}\hat{\mathbf{u}}_{B_{1,L}A_{2,R}} = r_p \begin{bmatrix} \cos\phi \\ \sin\phi \\ 0 \end{bmatrix} \times \begin{bmatrix} -\sin\phi \\ \cos\phi \\ 0 \end{bmatrix} = r_p \begin{bmatrix} 0 \\ 0 \\ \cos^2\phi + \sin^2\phi \end{bmatrix} = r_p\,\hat{\mathbf{k}}. \tag{45}
$$

The $\hat{\mathbf{k}}$ component collapses to 1 by the Pythagorean identity regardless of ϕ.
Projecting onto $\hat{\mathbf{k}}$.

$$
J_l = \left({}^{W}\mathbf{r}_{O_2A_{2,R}} \times {}^{W}\hat{\mathbf{u}}\right)^{\top}\hat{\mathbf{k}} = \left(r_p\,\hat{\mathbf{k}}\right)^{\top}\hat{\mathbf{k}} = r_p\,\underbrace{\hat{\mathbf{k}}^{\top}\hat{\mathbf{k}}}_{=1} = r_p. \tag{46}
$$

$$
\boxed{J_l = r_p} \tag{47}
$$

This is the belt (pitch-circle) formula: the effective moment arm of a cable tangent to a pulley of pitch radius r_p equals r_p for every cable angle ϕ, which is exactly the constant mapping encoded in the actuation matrix $\boldsymbol{A} = \mathrm{diag}(1, r_p)$ from eq. (8).

Geometric intuition. The cross-product magnitude $|\mathbf{r} \times \hat{\mathbf{u}}| = |\mathbf{r}||\hat{\mathbf{u}}|\sin\alpha$, where α is the angle between the lever arm and cable direction. Tangency forces $\alpha = 90$; hence $\sin\alpha = 1$ and the magnitude locks to $|\mathbf{r}| = r_p$ regardless of joint angle q_2. The spatial-velocity framework makes this structure manifest: it is built into the cross-product term of every row of the structure matrix V.

Corollary 12.1 (Attachment-point invariance)**.** Moving the cable attachment point $A_{2,R}$ to any other location on the same pitch circle — i.e. changing the angle ϕ while keeping $|{}^{W}\mathbf{r}_{O_2A_{2,R}}| = r_p$ — does **not** change J_l. Equivalently, $J_l = r_p$ is independent of (i) the joint angle q_2, and (ii) the circumferential position of $A_{2,R}$ on the pulley.

Proof. By eq. (45), the cross product evaluates to $r_p(\cos^2\phi + \sin^2\phi)\,\hat{\mathbf{k}} = r_p\,\hat{\mathbf{k}}$. The Pythagorean identity eliminates ϕ completely, so the projection onto $\hat{\mathbf{k}}$ in eq. (46) gives $J_l = r_p$ for all ϕ. □

Consequence for re-design. The three cases below follow directly from the corollary:

- *Different ϕ, same r_p*: rotating or repositioning $A_{2,R}$ around the pulley centre leaves J_l unchanged. Only the cable path geometry (tangent angle, wrap arc) changes.
- *Different q_2*: as the link rotates, $A_{2,R}$ moves in world frame but remains on the pitch circle. J_l stays constant — this is why the actuation matrix $\boldsymbol{A} = \operatorname{diag}(1, r_p)$ is pose-independent.
- *Different r_p*: swapping to a pulley of radius r_p' scales $J_l \to r_p'$ and proportionally rescales the entire hybrid-Jacobian column $\boldsymbol{J}_h^{(2)} = \boldsymbol{J}^{(2)}/r_p$.

13 Concrete Velocity Mappings for the CupManipulatorTendon

This section collects, in one place, all three velocity-level mappings that are used in the Python implementation:

1. $\dot{\boldsymbol{q}} \;\to\; \dot{l}$ (cable length rate, single cable to joint 2)
2. $\dot{\boldsymbol{q}} \;\to\; \dot{\boldsymbol{p}}$ (end-effector Cartesian velocity)
3. $\dot{l},\ \dot{q}_1 \;\to\; \dot{\boldsymbol{p}}$ (actuation-space to EE velocity)

Numerical values used throughout:

$$L_1 = 0.34247 \text{ m}, \quad L_2 = 0.19000 \text{ m}, \quad r_p = 0.04775 \text{ m}.$$

13.1 Joint-Velocity to Cable-Length Rates

Joint 2 is driven by **two cables**, green (G) and red (R), which wrap on the big pulley from *opposite sides* (see section 13.5). A positive $\dot{q}_2$ simultaneously *pays out* the green cable and *takes up* the red cable. From the belt formula eq. (47) applied to each cable:

$$\boxed{\dot{l}_G = +r_p\,\dot{q}_2, \qquad \dot{l}_R = -r_p\,\dot{q}_2.} \tag{48}$$

The two rates are not independent: they satisfy the *antagonistic constraint* $\dot{l}_G + \dot{l}_R = 0$ at all times (eq. (56)). Therefore only **one scalar** $\dot{l} \triangleq \dot{l}_G = -\dot{l}_R$ is needed to fully describe the cable kinematics of joint 2.

Dependencies: $\dot{l}_G$ and $\dot{l}_R$ depend *only* on $\dot{q}_2$; they are independent of q_1, q_2, and $\dot{q}_1$.

In vector form for the full joint-velocity vector $\dot{\boldsymbol{q}} = [\dot{q}_1,\ \dot{q}_2]^\top$:

$$\boldsymbol{l}\begin{pmatrix}\dot{l}_G \\ \dot{l}_R\end{pmatrix} = \underbrace{r_p \begin{bmatrix} 0 & +1 \\ 0 & -1 \end{bmatrix}}_{\boldsymbol{J}_l \,\in\, \mathbb{R}^{2\times 2}} \begin{bmatrix}\dot{q}_1 \\ \dot{q}_2\end{bmatrix}. \tag{49}$$

Because the rows are linearly dependent (rank 1), the equivalent *scalar* row Jacobian using only $\dot{l} \triangleq \dot{l}_G$ is:

$$\dot{l}_G = \underbrace{\begin{bmatrix} 0 & r_p \end{bmatrix}}_{\boldsymbol{J}_{l,G} \,\in\, \mathbb{R}^{1\times 2}} \begin{bmatrix}\dot{q}_1 \\ \dot{q}_2\end{bmatrix} = 0.04775\,\dot{q}_2 \quad [\text{m/s per rad/s}]. \tag{50}$$

The hybrid Jacobian (section 13.3) uses $\dot{l} \equiv \dot{l}_G$ as the second actuation-space coordinate; $\dot{l}_R$ is then implicit via $\dot{l}_R = -\dot{l}_G$.

13.2 Joint-Velocity to End-Effector Velocity

From the geometric Jacobian eq. (2):

$$\boxed{\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \end{bmatrix} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}} \tag{51}$$

Expanding the two rows explicitly:

$$\dot{p}_x = -(L_1 s_1 + L_2 s_{12})\,\dot{q}_1 \;-\; L_2 s_{12}\,\dot{q}_2, \tag{52}$$

$$\dot{p}_y = (L_1 c_1 + L_2 c_{12})\,\dot{q}_1 \;+\; L_2 c_{12}\,\dot{q}_2. \tag{53}$$

Observation. $\dot{q}_1$ controls both $\dot{p}_x$ and $\dot{p}_y$ via the full arm reach $(L_1 + L_2)$, whereas $\dot{q}_2$ controls them only via the distal link L_2. Near $q_2 = 0$ (arm straight), the second column of $\boldsymbol{J}$ becomes parallel to the first, causing $\det \boldsymbol{J} = L_1 L_2 \sin q_2 \to 0$.

13.3 Actuation-Space to EE Velocity (Hybrid Form)

The hybrid Jacobian from eq. (14) maps the *actuation-space velocity* $\dot{\boldsymbol{u}}_v = [\dot{q}_1,\, \dot{l}_G]^\top$ to EE velocity (recall $\dot{l}_G = -\dot{l}_R$ so $\dot{l}_R$ adds no new information):

$$\dot{\boldsymbol{p}} = \boldsymbol{J}_h\,\dot{\boldsymbol{u}}_v \quad \Longleftrightarrow \quad \begin{bmatrix} \dot{p}_x \\ \dot{p}_y \end{bmatrix} = \underbrace{\begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -\dfrac{L_2 s_{12}}{r_p} \\ L_1 c_1 + L_2 c_{12} & \dfrac{L_2 c_{12}}{r_p} \end{bmatrix}}_{\boldsymbol{J}_h(\boldsymbol{q})} \begin{bmatrix} \dot{q}_1 \\ \dot{l}_G \end{bmatrix}. \tag{54}$$

Numerically, $L_2/r_p = 0.190/0.04775 \approx 3.979$, so the cable column amplifies EE velocity by a factor of ≈ 4 compared to a direct-drive joint of the same link length.

13.4 Complete Mapping Summary

Mapping	Equation	Notes
$\dot{q}_2 \to \dot{l}_G,\, \dot{l}_R$	$\dot{l}_G = +r_p\dot{q}_2,\; \dot{l}_R = -r_p\dot{q}_2$	Antagonistic pair; $\dot{l}_G + \dot{l}_R = 0$
$\dot{q}_2 \to \dot{l}_G$ (scalar)	$\dot{l}_G = r_p\,\dot{q}_2$	Used as the independent cable rate
$\dot{\boldsymbol{q}} \to \dot{\boldsymbol{p}}$ (joint space)	$\dot{\boldsymbol{p}} = \boldsymbol{J}(\boldsymbol{q})\,\dot{\boldsymbol{q}}$	Standard geometric Jacobian eq. (2)
$\dot{\boldsymbol{q}} \to \dot{\boldsymbol{p}}$ (actuation space)	$\dot{\boldsymbol{p}} = \boldsymbol{J}_h(\boldsymbol{q})\,\dot{\boldsymbol{u}}_v$	$\dot{\boldsymbol{u}}_v = [\dot{q}_1,\, \dot{l}_G]^\top$; $\dot{l}_R$ implicit
$\dot{\boldsymbol{p}} \to \dot{\boldsymbol{q}}$ (velocity IK)	$\dot{\boldsymbol{q}} = \boldsymbol{J}^\dagger\dot{\boldsymbol{p}}$	Damped pseudo-inverse, singular near $q_2 = 0$
$\dot{\boldsymbol{p}} \to \dot{\boldsymbol{u}}_v$ (hybrid vel IK)	$\dot{\boldsymbol{u}}_v = \boldsymbol{J}_h^\dagger\dot{\boldsymbol{p}}$	Used in `compute_velocity_ik()`

Python implementation. In `robots/cup_manipulator_tendon.py`:

- `compute_ik_analytical(plant, target_xy, ...)` solves $\dot{\boldsymbol{q}} \to \boldsymbol{p}$ by the closed-form 2R formula (preferred).
- `compute_velocity_ik(plant, context, ee_velocity_xy, damping)` computes $\dot{\boldsymbol{u}}_v = \boldsymbol{J}_h^{\dagger}\,\dot{\boldsymbol{p}}$ via the damped pseudo-inverse. $\dot{l}_G$ is then recovered as $\dot{l}_G = (\dot{\boldsymbol{u}}_v)_2$, $\dot{q}_2 = \dot{l}_G / r_p$, and $\dot{l}_R = -\dot{l}_G$ (antagonistic constraint).

13.5 Antagonistic Cable Pair

Joint 2 is driven by **two cables that act in opposition** — analogous to agonist/antagonist muscle pairs in biology. Each cable can only *pull* (tension ≥ 0); the antagonistic arrangement allows bidirectional torque on a passive revolute joint.

Physical routing (from `CableRig`).

Cable	Colour	Route through pulleys
Agonist	**Green**	$\text{Start}_R \to \text{Drive}_{B_R} \xrightarrow{\text{wrap}} \text{IdlerR} \xrightarrow{\text{wrap}} \text{BigPulley}_{A_L \to B_L} \to \text{EndL}$
Antagonist	**Red**	$\text{Start}_L \to \text{Drive}_{B_L} \xrightarrow{\text{wrap}} \text{IdlerL} \xrightarrow{\text{wrap}} \text{BigPulley}_{A_R \to B_R} \to \text{EndR}$

The two cables enter the *big pulley* (rigidly attached to link 2, body `link2_tendon`) from **opposite sides**: green from the left (A_L) and red from the right (A_R). This means they wrap in opposing angular senses, producing opposite torques on joint 2.

Cable-rate kinematics of the antagonistic pair. Let l_G and l_R denote the *free lengths* (drive-to-anchor lengths) of the green and red cables respectively. Since the wrap arcs on the big pulley are complementary, a rotation $\dot{q}_2$ simultaneously pays out one cable and takes up the other:

$$\dot{l}_G = +r_p\,\dot{q}_2, \qquad \dot{l}_R = -r_p\,\dot{q}_2. \tag{55}$$

Summing the two rates gives the *antagonistic constraint*:

$$\boxed{\dot{l}_G + \dot{l}_R = 0} \tag{56}$$

This is the cable equivalent of muscle incompressibility: whatever length the agonist gains, the antagonist gives up.

In matrix form for $\dot{\boldsymbol{l}} = [\dot{l}_G,\, \dot{l}_R]^\top$:

$$\dot{\boldsymbol{l}} = \underbrace{r_p \begin{bmatrix} 0 & +1 \\ 0 & -1 \end{bmatrix}}_{\boldsymbol{J}_l^{(2)} \in \mathbb{R}^{2\times 2}} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}. \tag{57}$$

Net torque on joint 2. Each cable exerts a tension force on the big pulley with a moment arm of r_p. Because the cables wrap in opposite senses, green tension $f_G \geq 0$ produces positive torque and red tension $f_R \geq 0$ produces negative torque (or vice versa, depending on the positive-q_2 convention):

$$\boxed{\tau_2 = r_p\,(f_G - f_R), \qquad f_G,\, f_R \geq 0.} \tag{58}$$

This matches the structure-matrix transposition: the wrench on joint 2 from the cable-force vector $_\mathbf{c} = [\mathbf{f_G},\, \mathbf{f_R}]^\top$ is $\boldsymbol{\tau} = (\boldsymbol{J}_l^{(2)})^\top{}_\mathbf{c}$, giving $\tau_2 = r_p(f_G - f_R)$ and $\tau_1 = 0$ (cables do not act on joint 1).

Drive-motor coupling. The drive pulley (on the base, body `link1_base`) rotates by θ_{drv}. Its two exits feed the green and red cables with *equal and opposite* length changes (the drive pulley is also symmetric):

$$\dot{l}_G^{\text{drv}} = -r_{\text{drv}}\,\dot{\theta}_{\text{drv}}, \qquad \dot{l}_R^{\text{drv}} = +r_{\text{drv}}\,\dot{\theta}_{\text{drv}}, \tag{59}$$

where $r_{\text{drv}} = r_p$ (same HTD 5M tooth pitch). The net free length of each cable is determined by the *difference* of big-pulley and drive-pulley contributions. In the implementation, the drive motor is the sole control input u_2 for joint 2; the antagonistic constraint ensures that setting $\dot{\theta}_{\text{drv}}$ is sufficient to specify both $\dot{l}_G$ and $\dot{l}_R$ simultaneously.

Admissible-force cone. Since both cable tensions must be non-negative, the net torque is bounded:

$$\tau_2 \in [-r_p f_{\max},\ +r_p f_{\max}], \tag{60}$$

but the pair can also generate a *controlled stiffness*: choosing $f_G = f_R = f_{\text{co}} > 0$ (co-contraction) produces $\tau_2 = 0$ while increasing the effective joint stiffness. This is the cable analogue of muscle co-activation.

References

[1] A. Pott and T. Bruckmann, *Cable-Driven Parallel Robots: Theory and Application*, Springer Tracts in Advanced Robotics, 2018.